DO NOW — RECALL

SQL SELECT, JOIN and nested SELECT

Name: _____ Class: _____ Date: _____

1. What is the purpose of an INNER JOIN?

2. Which relational key connects two tables?

3. What does WHERE do?

4. Binary search for 9 in sorted [1,3,5,7,9,11,13]. Show midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Combines matching rows.

2. Model answer

Foreign key referencing primary key.

3. Model answer

Filters results according to a predicate.

4. Model answer

low0 high6 mid3→7 then low4 high6 mid5→11 then high4 mid4→9, found index4.

Vocabulary — know these terms

SELECT — Choose fields for output.

INNER JOIN — Combine matching rows.

nested SELECT — Query inside another query.

IN — Compare value against returned set.

WHERE — Filter rows.

NEW LEARNING

Read, annotate and explain

Understand

SELECT names output columns, FROM identifies input tables and WHERE restricts rows using a Boolean predicate.

Trace

INNER JOIN combines rows matching a join condition. A nested SELECT can return a set of keys used by an outer IN condition.

Explain

Both example queries return Ada and Ben, but joining is more flexible when fields from both tables are required.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Write SQL for student names where Score.Mark>80.

Predict

Only Ben qualifies because score 82 is the sole result above 80.

Check the answer

Explain which rule makes this result correct.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write SELECT to show all Name values in Student sorted by Name ascending. [4]

Question 2

Write and trace a join returning Name and Mark for scores >=70. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write a nested SELECT for students scoring above 75. Explain result. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

INNER JOIN results can be the same as a nested key SELECT for a particular data set.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.